

The Causal Effect of Special Elections on Media Internal Pluralism: A Historical Investigation

POLI 228 – Historical Political Economy

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Abstract. This paper investigates the causal effect of electoral campaign intensity on the internal pluralism of newspaper coverage using a stacked difference-in-differences design. Exploiting deaths of U.S. House members as exogenous shocks to the political agenda, I analyze over 1.3 million newspaper articles from six outlets across 100 years of American history. First, I find that special election campaigns do not increase the number of political articles published. Second, I find that special election campaigns increase partisan slant in headline coverage, with no evidence for heterogeneous effects over time. These results suggest that electoral competition reduces the internal pluralism of political information available to voters, with implications for theories of democratic accountability and media quality. However, the estimated slant effect is not robust to more demanding specifications, such as a donut difference-in-differences design.

Keywords: media pluralism, special elections, difference-in-differences, partisan bias, political polarization

No democracy can afford market failure in
this sector.
— Jürgen Habermas, (2010)

1. Introduction

Classical models of political accountability are based on the assumption of full, or at least sufficiently rich, information (Ashworth, De Mesquita, and Friedenberg 2017; Downs 1957; Fiorina 1978). Voters, however, do not perceive politics directly, but rather through the mediation of the media (Iyengar and Kinder 2010; McCombs and Shaw 1972; J. R. Zaller 1992; J. Zaller 1999). I argue that the quality of the media environment feeds back into the quality of the information available to voters and thus makes accountability a function of media quality (Snyder and Strömberg 2010; Strömberg 2004). Changes in the media environment may therefore constitute part of the explanation for democratic erosion as well as rising political polarization over time (Lelkes, Sood, and Iyengar 2017; Prior 2007). Whereas the traditional accountability literature primarily emphasizes the availability of political information to voters (Ashworth 2012; Strömberg 2015), the quality of that information has received comparatively less attention and has only recently become central to debates on fake news and misinformation (Allcott and Gentzkow 2017). With this paper, I do not contribute to the debate on false information; rather, I define media quality as partisan bias in news coverage (Gentzkow and Shapiro 2010).

The quality of information is especially important in the period before an election, when voters make their electoral decisions. In line with Gelman and King (1993), I argue that the campaign period is a time of heightened political intensity (Campbell et al. 1960; Lazarsfeld, Berelson, and Gaudet 2021; J. R. Zaller 1992) in which, on the one hand, voters have a particularly strong demand for political information that aligns with their prior political views (Stroud 2011). At the same time, election campaigns generate a particularly high supply of political information (Iyengar and Simon 2000) produced by parties and political elites, which is then taken up by the media (Druckman, Peterson, and Slothuus 2013). Together, these dynamics cause the quality of media coverage to decline during campaign periods, conditional on the level of political intensity.

For this reason, this paper asks *what effect political competition intensity has on media coverage*, focusing on the United States because of its pronounced and relatively stable media market in one of the world's oldest democracies (Petrova 2011; Snyder and Strömberg 2010). Adopting a historical political economy perspective, I examine a period spanning 100 years of U.S. history. This long time horizon allows me to identify a sufficient number of special elections, which I use as exogenous shocks to the media agenda in order to identify causal effects

(McCombs and Shaw 1972).

This study makes several contributions. First, to the best of my knowledge, it is the first study to systematically compile historical data on special elections. Second, I show the causal effect of campaign intensity on media behavior, examining both the number of political news articles and, in a second step, media quality as measured by partisan slant. The paper thus contributes to the literature on media quality and journalistic norms in historical perspective. In a broader context, it also contributes to the understanding of democratic backsliding and political polarization.

This study proceeds as follows. I review existing theories on the effect of political intensity on domestic media pluralism, identify several political-economic micro-mechanisms, and derive three hypotheses. I test these hypotheses using descriptive statistics, event-study plots, and stacked difference-in-differences models. I show that while special election campaigns do not affect the number of political articles published, they have a positive effect on the partisan slant of newspaper coverage. Furthermore, I find no evidence for a time-conditional effect of special election campaigns. In robustness tests, however, I show that the slant effect is not robust to more demanding specifications such as a donut-difference-in-differences design. Finally, I discuss the results, highlight limitations, and outline directions for future research.

2. Theory

2.1 Media, Function, and Pluralism

Media report selectively and, in doing so, structure both public opinion and the political agenda. From this perspective, the media are not merely transmitters, but political institutions (Cook 2005) whose organizational forms and market logics vary systematically (Baumgartner and Jones 2009). In their comparative analysis of media systems, Hallin and Mancini (2004) identify differences along the dimensions of political parallelism, journalistic professionalism, state intervention, and market structures. In systems with high political parallelism, pluralism is often organized through differentiation across outlets, while a more coherent line tends to dominate within individual media organizations; in more professionalized or public-service-oriented environments, pluralism is more likely to be institutionalized within individual media outlets (Blumler and Gurevitch 1995; Chadwick 2017; Hallin and Mancini 2004).

The political economy literature points to a trade-off between internal pluralism and pluralism across media outlets (Gentzkow and Shapiro 2010; **mul-**

lainathan?). In media markets with strongly competitive structures, and the correspondingly higher degree of pluralism across outlets that often accompanies them, the internal pluralism of content tends to decline (Gentzkow and Shapiro 2006; Gentzkow and Shapiro 2010; mullainathan?). Outlets differentiate themselves as brands or niches; internal heterogeneity weakens branding, and internal pluralism therefore declines. In intense campaigns, competition for attention increases, so this effect should become even stronger. An exception in such media markets may be state media, which typically carry a political mandate to maintain internal pluralism (Hallin and Mancini 2004).

I argue that the internal pluralism of media is of decisive importance for the accountability of political elites and parties. Stroud (2011) show that voters strongly self-select into media outlets on the basis of their partisanship (Iyengar and Hahn 2009; Prior 2007). Pluralistic media consumption is therefore rare. Newspapers often emerge from a tradition of strong ties to political parties (Hallin and Mancini 2004; Kaplan 2003) and, for this reason, already tend to exhibit lower internal pluralism than other media forms, in which, traditionally, broadcast licenses were more limited and the state intervened more strongly in the media market (Hallin and Mancini 2004). The U.S. media market is known for its limited state intervention and strong competition (Hallin and Mancini 2004; Hamilton 2004). For this reason, I argue that, especially in the United States, the pluralism of the information voters consume depends on the internal pluralism of the media.

2.2 Mechanismen

Media behavior is shaped to a large extent by three external forces: state intervention, professional journalistic norms, and economic constraints (Cook 2005; Hallin and Mancini 2004; Hamilton 2004). I argue that, by conducting a case study within a single democratic country, I hold the first two factors constant by design, so that media behavior is driven primarily by economic constraints.

2.2.1 Demand Side

Campaigns increase identity salience and emotional involvement; as a result, motivated reasoning intensifies and the demand for congenial information rises (Taber and Lodge 2006), while media outlets respond in profit- and audience-maximizing ways by adopting a more coherent “house line.” Voters prefer news that is consistent with their affectively grounded attitudes in order to reduce cognitive dissonance, and they search for and interpret information selectively (Gentzkow and Shapiro 2010; Lodge and Taber 2013). Huddy, Mason, and Aarøe (2015) show

that campaign events activate strong emotions tied to partisan identity, thereby reinforcing the effect of affectively grounded attitudes. This makes social sorting on the basis of partisanship more likely during periods of high salience (Iyengar and Hahn 2009; Mason 2016, 2018; Messing and Westwood 2014).

Media are economic actors and seek, above all, to maximize their audience and, with it, their profits. Gentzkow and Shapiro (2010) show that outlets respond systematically to consumer demand, while (mullainathan?) model confirmatory preferences and show that competition can shift an outlet's slant toward the prior beliefs of its readers (Hamilton 2004).

2.2.2 Supply Side

Campaigns constantly produce “ready-made” inputs such as events, PR, and talking points. Under time pressure and tight production cycles, newsrooms rely on standardized templates such as the horse-race frame, strategy/game framing, and scandal coverage (Aalberg, Strömbäck, and De Vreese 2012; Cappella and Jamieson 1997; Patterson 2011); this crowds out more heterogeneous and time-intensive perspectives. Political actors have different strategic means of shaping the media agenda, the most influential of which are framing (Aalberg, Strömbäck, and De Vreese 2012; Vreese 2005), agenda-setting (McCombs and Shaw 1972; McCombs and Valenzuela 2021), and priming (Iyengar and Kinder 2010). During campaigns, messages within political camps converge and become more tightly coordinated (Druckman, Peterson, and Slothuus 2013; J. R. Zaller 1992).

Due to resource constraints in news production, above all limited time, media rely on these prefabricated messages provided by political actors (Gans 2004; Shoemaker and Vos 2009). This effect is further intensified during campaign periods because the demand for political news rises, while the volume and complexity of political events also increase (Gans 2004; Shoemaker and Vos 2009).

2.2.3 Strategic Alliances

When access to candidates, surrogates, or embargoed information is central, access itself can function as a disciplinary instrument; media internalize this logic through self-censorship, source dependence, and indexing. Bennett (1990) describes how the range of legitimate views in news discourse is often “indexed” to official or elite debate; when elites control access, this narrows the range of views that enters coverage. This effect is further reinforced by journalistic routines governing the use of official sources, with the result that dissenting voices are screened out (Bennett,

Lawrence, and Livingston 2008; Cottle 2000; Gans 2004).

Competing theories instead assume that the increased demand for information during elections, uncertainty about the winner and the future policy direction (Downs 1957; Gentzkow and Shapiro 2006; Huddy, Mason, and Aarøe 2015), and also the stronger scrutiny of the media by watchdog institutions alongside self-censorship rooted in professional role conceptions (Schudson 2001; Tuchman 1972) lead to higher internal pluralism. In addition, this literature argues that outlets become less dependent on their core audience (Fowler 2022) and have more material available for their coverage (Dimitrova and Strömbäck 2012).

Based on that I derive three hypotheses:

H1: Campaign periods increase the demand for political news and therefore increase the number of articles published by the outlets.

H2: Campaign periods reduce the internal pluralism of media coverage, due to the increase in the demand for congenial information.

H3: The negative effects of campaign intensity increase over time as audience polarization, and thus demand-side polarization, rises.

3. Design

The analysis of media effects and effects on the media is subject to the fundamental problem of endogeneity. The literature has extensively described the influence of the media on the political process (Iyengar and Kinder 2010; McCombs and Shaw 1972). These effects primarily include agenda-setting, priming, and framing. From a causal perspective, the question is therefore whether pluralism in the media declines as a result of campaign intensity, or whether campaign intensity increases as a result of strongly partisan media. To address this endogeneity problem (Gerber, Karlan, and Bergan 2009; Snyder and Strömberg 2010), I use special elections as shocks to the political agenda. I argue that, under certain conditions, special elections can satisfy the requirements of an exogenous shock.

3.1 Special Elections as an Exogenous Shock

Special elections are elections that are called, or may be called, when a seat in a U.S. legislature becomes vacant in order to fill it. In general, there is substantial variation in the institutional rules governing how a vacant seat is filled. In this study, I focus on the House of Representatives because the U.S. Constitution requires House vacancies to be filled by election rather than appointment. Article I, Section 2, Clause 4 provides that, when vacancies occur in a state's representation, the state executive "shall issue Writs of Election to fill such Vacancies." By contrast, Senate vacancies may often be filled temporarily by gubernatorial appoint-

ment under the Seventeenth Amendment, depending on state law. This makes the House institutionally especially well suited for studying vacancy-triggered special elections. When a seat becomes vacant close to the end of the term, however, no special election is often held, and the seat is instead filled at the next general election.

I argue that the occurrence of a special election is not, by itself, sufficient for it to serve as an exogenous shock. In principle, there is substantial variation in the reasons why a seat becomes vacant. The most common cases are resignation and death while in office. The reasons for resignation can be highly diverse, and historically it is often difficult to reconstruct the full motivations of politicians. I discuss three main reasons here: resignations due to political scandals, resignations for other political offices, and strategic resignations.

3.1.1 Resignations

Political scandals pose a problem for the operationalization of special elections as an exogenous shock, because they violate the “no co-treatment” assumption that is fundamental to causal identification. In such cases, the media are not concerned only with the election campaign itself, but also with the scandal that prompted the politician’s resignation.

Resignations for other political offices violate the assumption that the shock is unforeseen, and thus exogenous. Most politicians who run for another office, such as governor, senator, or a position in the federal administration, do not do so randomly, but instead prepare themselves and their political environment for this transition, which in such cases unfolds in a planned and structured manner. The same applies to strategic resignations, for example when a party uses a moment of high public support to engineer a generational transition rather than waiting for the next election. Overall, I argue that resignations are generally not suitable as exogenous shocks and are therefore not analyzed in this paper.

3.1.2 Death

Compared to resignations, deaths in office constitute a far more unforeseen shock to the political agenda. Yet, it is necessary to distinguish the manner in which the officeholder died. In the case of a person with a known illness, death is less unforeseen than in the case of someone who dies as a result of external violence, such as an accident or physical attack. Figure 1 shows the distribution of causes of death by category for members who died while serving in the House. The most

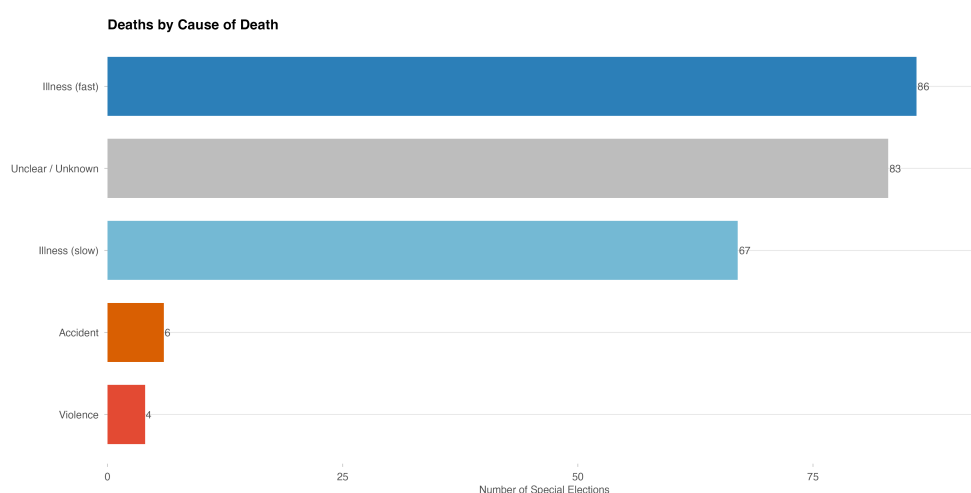


Figure 1: Distribution of causes of death for members of Congress who died in office, 1789–2020. Illness (fast) constitutes the primary identification category.

common cause is fast-onset illness, which I define as a non-chronic condition not known prior to the member taking office and from which the member typically died within six weeks. The second most common category, with 83 cases, consists of individuals whose cause of death is unclear or unknown. This is followed by chronic illness; accidents and violence constitute a small minority. In total, I identified 246 cases in which a special election was held due to the death of a House member; in 163 of these cases, I was able to identify and classify the cause of death.

3.2 Data

To test the derived hypotheses, I construct an original dataset that links special elections to media behavior. The dataset I compile is structured at the week-by-newspaper-outlet level. The observation period for special elections begins with the First Congress in 1789 and extends to the 2020 election. The observation period for the newspaper analysis is shorter due to data availability and covers the period from 1920 to 2020.

3.2.1 Special election Data

As the data foundation for special elections, I merge datasets on members of Congress with datasets on special elections. The primary data source is the *Biographical Directory of the United States Congress* (Bioguide). The dataset is structured at the legislator-by-Congress level and includes information on the beginning and end of representatives' terms, as well as, in cases where a term ends early, the reason for that early termination. However, the recorded reason for the

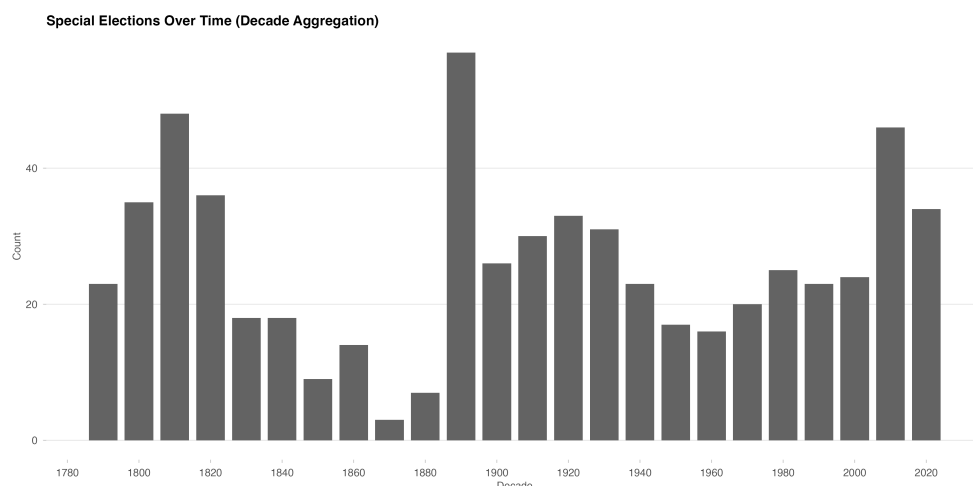


Figure 2: Number of special elections by cause of vacancy over time (1789–2020).

end of a term is fairly coarse and usually contains only information on whether the individual died or resigned. In addition, the dataset includes a biographical text with further information on each legislator. One challenge in the data is that the beginning of the term does not necessarily correspond to the time at which legislators were sworn in.

I combine the individual congressional data with data on special elections. Existing datasets on special elections, such as those by Hirano and Snyder, Jr. (2009), cover only a fraction of the full observation period. The only comprehensive dataset on special elections that extends back to the First Congress is available on Wikipedia (List of special elections to the United States House of Representatives 2026). Although the data quality of Wikipedia may reasonably be questioned, I justify its use in this project because no better source exists for the full period. I validate these data using information from the notes contained in Bioguide.

Figure 2 displays the number of special elections by decade. Special elections are not uniformly distributed over time; their frequency reflects historical and institutional variation, including changes in mortality, resignation patterns, political career incentives, and state-specific rules for filling House vacancies. Because the timing of replacement elections is regulated by state law, I expect regional variation.

Information on causes of death also comes from Wikipedia (List of members of the United States Congress who died in office (1900–1949) 2026). Wikipedia in turn lists *Memorial Addresses* from HathiTrust as the primary source. Again, I use the Bioguide information available to me to verify the recorded causes of

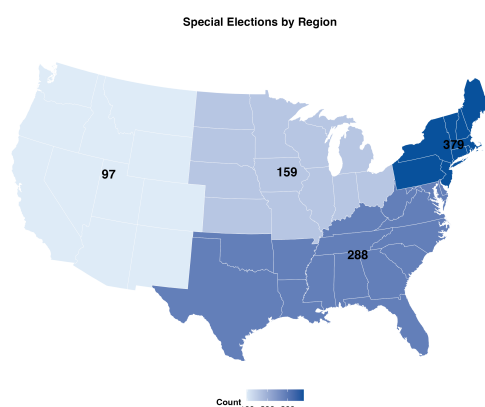


Figure 3: Geographic distribution of special elections by U.S. region (1789–2020).

death. I then classify causes of death by means of keyword matching into the categories illness (fast), illness (slow), accident, violence, and unclear/unknown. In later analyses, I distinguish between these different forms of death to ensure robustness and exogeneity.

I merge these datasets into a joint special election dataset that records, for each special election, who left Congress and who was elected as a replacement, including the relevant biographical information, the reason the special election occurred, and, if the original member of Congress died, the cause of death.

Figure 3 shows the distribution of special elections by U.S. region over the full observation period. This figure is confounded by the fact that states entered the Union at different historical moments and are therefore not observed for the same length of time. Earlier-admitted states, particularly in the Northeast and parts of the South, mechanically accumulate more special elections than later-admitted states in the West. Figure 4 shows the distribution of causes of death among House members by region. The same confound applies, but importantly, there is no evidence of systematic sorting of causes of death by region, for instance due to regional disease outbreaks or concentrated political violence.

3.2.2 Newspaper Data

The newspaper data come from ProQuest TDM Studio. TDM Studio provides access to a large number of newspaper archives, including historical newspapers collected and curated by ProQuest itself. For my research design, I consider only U.S. newspaper outlets for which historical newspaper articles are available. Given computational constraints, while still aiming to ensure geographic coverage, I select representative outlets from each region of the United States. I focus on newspapers with a long tradition in their respective regions and a reputation for journalistic

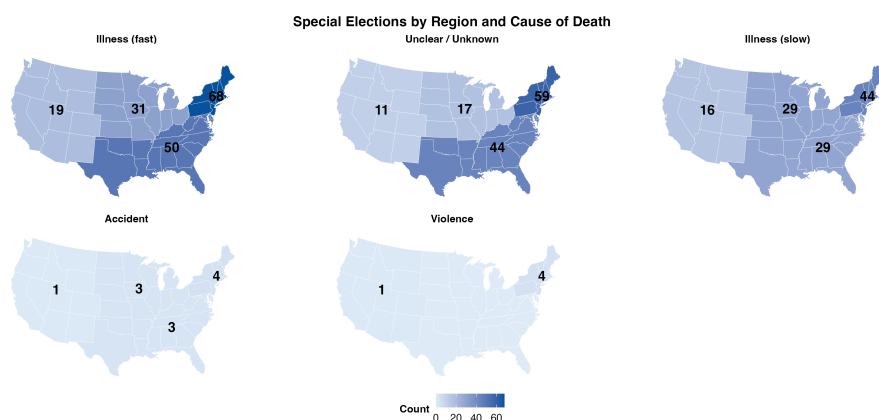


Figure 4: Geographic distribution of special elections by region and cause of death.

standards, so that no obvious biases are introduced into my models.

For the Northeast, I use historical newspaper articles from the *New York Times*. For the South, I draw on *The Atlanta Constitution* and *The Baltimore Sun*. For the Midwest, I use historical articles from the *Chicago Tribune*, and for the West, I use the *Los Angeles Times* and the *Los Angeles Sentinel*.

To make the number of articles manageable, I restrict the sample to English-language articles classified as “news” and containing political keywords such as *Congress*, *election*, or *campaign* in the full text (for the full search prompt, see Appendix A). After filtering, the dataset contains 1,347,370 articles.

To measure partisan tone, I use the large language model Llama 3.2, a 3-billion-parameter open-source model, to classify the partisan slant of each article headline. For each headline, the model is prompted to assign a directional slant score: positive values indicate Republican-leaning coverage, negative values indicate Democratic-leaning coverage, and zero indicates neutral framing. The prompt instructs the model to evaluate whether the headline implicitly favors one party’s positions, candidates, or narrative frame over the other’s (for the full LLM prompt, see Appendix A).

The resulting headline-level slant scores are then aggregated to the outlet-week level by computing the mean score across all articles published by a given outlet in a given week. This outlet-week mean constitutes the primary outcome variable, *slant*, used in the event-study and difference-in-differences specifications.

As a secondary outcome measuring the *total slant* of coverage rather than its direction I use the absolute value of each headline-level score, $|\text{slant}|$, averaged within outlet-weeks. A higher absolute score indicates that articles cluster at one partisan extreme, regardless of direction, and thus captures internal homogeneity of coverage.

This measurement approach has several advantages. First, it scales to the full corpus of 1.3 million articles without requiring manual coding. Second, because the model evaluates headlines rather than full article text, it focuses on the most salient editorial signal visible to readers. Third, aggregating to the outlet-week level smooths idiosyncratic noise in individual headline classifications and aligns with the panel structure of the design. A limitation is that LLM-based classification may introduce systematic errors for historically distant language; I address this by reporting results separately for pre- and post-1970 subsamples.

3.3 Identification Strategy and Threats to Identification

For identification, I employ a stacked difference-in-differences (DiD) design combined with event-study plots to assess dynamic treatment effects (Angrist and Pischke 2008; Roth et al. 2023). The stacked DiD approach constructs a separate clean comparison dataset for each event (i.e., each special election), in which the treated region is matched against all not-yet-treated or never-treated control regions observed in the same event window. These event-specific datasets are then stacked and estimated jointly. This avoids the negative-weighting problem that arises in two-way fixed-effects estimators with heterogeneous treatment timing (Roth et al. 2023).

The estimating equation takes the form:

$$Y_{irt} = \sum_{\tau \neq -1} \beta_{\tau} \cdot \mathbf{1}[\text{EventTime}_{it} = \tau] \times \text{Treated}_{ir} + \delta_i + \gamma_t + \varepsilon_{irt},$$

where Y_{irt} is the outcome (partisan slant or article volume) for outlet i in event-specific stack r at relative time t . Treated_{ir} equals one if outlet i is located in the region where event r took place. δ_i are event fixed effects and γ_t are event-time fixed effects, both estimated within each event stack. Standard errors are clustered at the event level to account for within-event correlation across outlets and weeks.

The coefficients β_{τ} for $\tau < -1$ constitute a pre-trend test: under the parallel-trends assumption, these should be jointly indistinguishable from zero. The post-period coefficients ($\tau \geq 0$) capture the dynamic treatment effect over the campaign window. The pooled average treatment effect on the treated (ATT) is computed

as the weighted average of post-period β_τ estimates.

I use the stacked DiD rather than a simple pooled two-way fixed-effects specification for two reasons. First, the stacked design explicitly accounts for the fact that control units may themselves be treated in other event stacks, which would contaminate a naive estimator. Second, restricting each comparison to a well-defined event window of 25 weeks (12 pre-treatment + 1 reference + 12 post-treatment) for the volume analysis (*H1*) and 13 weeks for the tone analysis (*H2* and *H3*) respectively, prevents contamination across temporally overlapping events.

Because I do not observe newspaper circulation directly, the geographic effects can only be proxied. I therefore assign outlets to U.S. regions as the coarsest feasible level of geographic clustering. A region is treated if it is in a vacancy period preceding a special election. The remaining regions in the same time period serve as the control group.

The parallel-trends assumption is plausible given the exogenous nature of the treatment: the timing of a House member's death is not strategically chosen by either the outlet or political actors. During the campaign period, there may be additional influences on the media environment, such as political advertising by parties, candidates, or interest groups (Bode et al. 2016), but these are all causally downstream of the vacancy event and therefore form part of the reduced-form effect I seek to estimate.

To assess the robustness of the parallel-trends assumption beyond the visual pre-trend test, I apply the sensitivity analysis of Roth et al. (2023) (HonestDiD). This method characterizes the minimum deviation from parallel trends, expressed as a multiple $\bar{M}$ of the largest observed pre-trend, that would be required to overturn the post-period estimates. Results of this analysis for each main specification are presented in the appendix.

Figure 5 displays vacancy periods and the timing of special elections across regions over the full observation period and is intended to illustrate the research design. Each blue-shaded interval denotes a vacancy period preceding a special election, which is indicated by a red vertical line. In the empirical framework outlined here, this vacancy period defines the treatment period. The 12 weeks prior to the onset of the vacancy period are used as the pre-treatment window. Regions that are not treated during a given period serve as the control group for both the difference-in-differences specification and the event-study design.

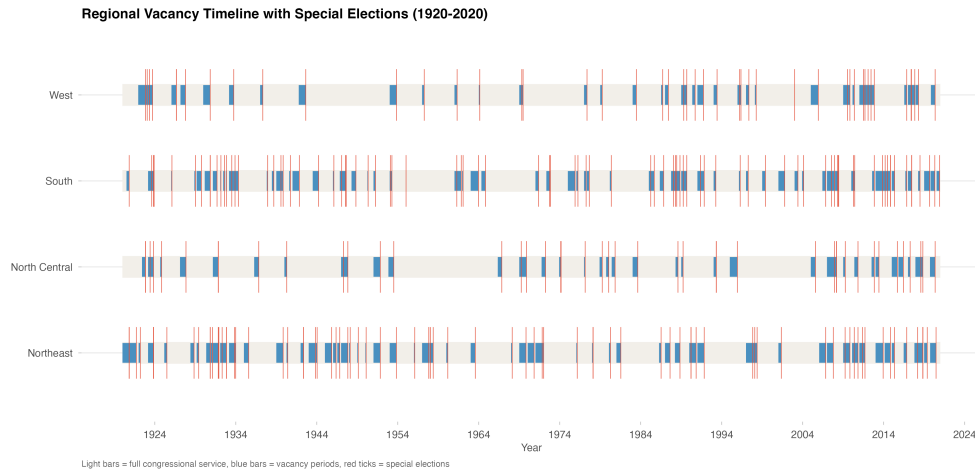


Figure 5: Vacancy periods and special elections by U.S. region, 1920 - 2020 .

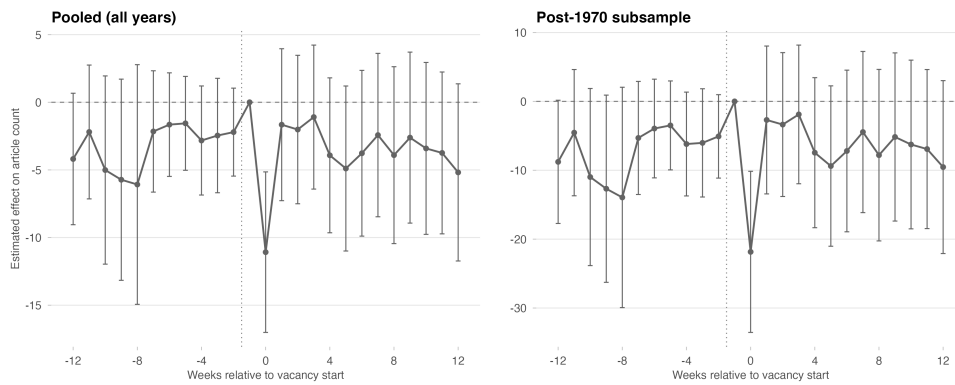


Figure 6: Event study for article volume. Left: pooled sample; right: post-1970 subsample. Error bars show 95% CI clustered by event.

4. Results

4.1 Article Volume

I first examine whether a special election affects the *volume* of political news coverage, that is, the number of articles published per outlet-week (*H1*). If the outlined mechanism holds, I expect to observe an increase in the number of political articles during the campaign period. Figure 7 shows the full event-study trajectories for article volume in the pooled and post-1970 samples. I distinguish between these two samples because the literature identifies the early 1970s as the beginning of a period of rising partisan polarization in American public opinion (Highton and Kam 2011; Layman, Carsey, and Horowitz 2006). If audience polarization amplifies demand-side pressure on outlets to adopt partisan framing, treatment effects should be larger in the post-1970 period. Pre-period coefficients are flat and jointly insignificant, supporting the parallel-trends assumption. Post-period coefficients

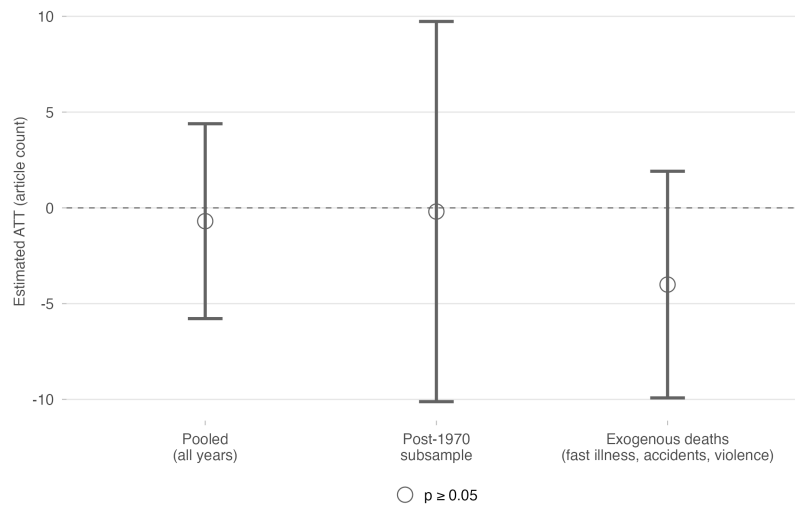


Figure 7: ATT estimates for article volume ($n_articles$). Stacked DiD, 95% CI clustered by event. None of the specifications yield a statistically significant effect.

are also close to zero and statistically indistinguishable from zero throughout the campaign window. This null result is consistent with the view that special elections shift the partisan framing of coverage without expanding its total volume.

Figure 8 displays the ATT estimates for article volume across the pooled sample, the post-1970 subsample, and the exogenous-deaths subsample. None of the specifications yields a statistically significant effect of the campaign period on article volume. This indicates that, even if the proposed mechanism holds, newspapers do not respond to increased demand for political news by expanding their supply. The exogenous-deaths specification even suggests a negative point estimate, though it too is imprecisely estimated. I therefore fail to find support for *H1* and reject it.

4.2 Partisan Tone

Having established a null effect on article volume, I turn to partisan tone. Figure 9 shows ATT estimates for headline slant across four specifications: the pooled sample, the post-1970 subsample, the exogenous-deaths subsample (fast illness, accidents, and violence combined), and the illness-slow subsample as a placebo comparison. The first two specifications represent the primary estimates; the exogenous-deaths specification provides the cleanest causal identification by restricting to the most unforeseen causes of vacancy; the illness-slow specification serves as a within-deaths placebo, since slow illnesses may violate the exogeneity assumption due to anticipation effects, so that I cannot rule out endogeneity problems.

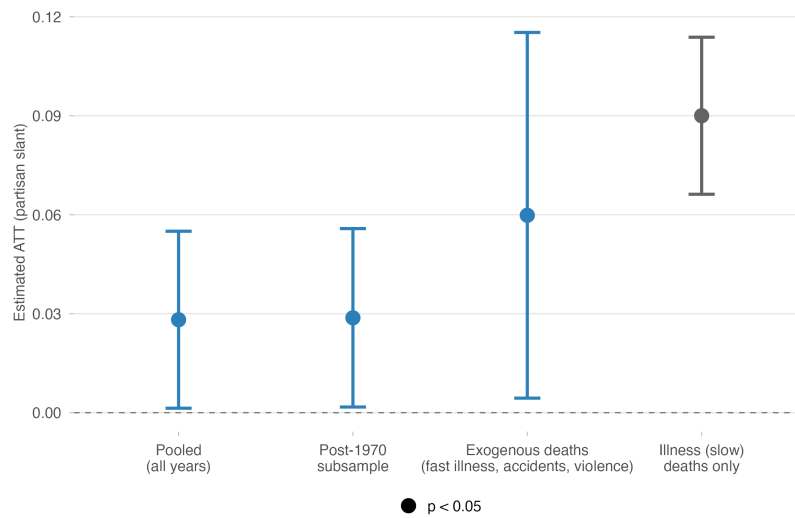


Figure 8: ATT estimates for partisan slant. Stacked DiD, 95% CI clustered by event. Blue indicates exogenous specifications (pooled, post-1970, exogenous deaths); grey indicates the illness-slow placebo. Filled circles denote statistical significance at the 5% level.

All four specifications yield statistically significant effects at the 95% confidence level. The pooled and post-1970 models show an average treatment effect of approximately 0.03 points on the slant scale, where positive values indicate Republican-leaning coverage and negative values indicate Democratic-leaning coverage. The pooled and post-1970 point estimates are nearly identical, providing no evidence of a temporal trend in line with *H3*. A pre-1970 model could not be estimated reliably due to the small number of observations and the absence of stable pre-trends.

The exogenous-deaths specification yields an effect approximately twice as large, which is consistent with the interpretation that non-exogenous vacancies attenuate the average estimated effect, for instance because strategic resignations lack the abrupt competitive shock that drives media polarization. The illness-slow specification, shown in grey, serves as a placebo comparison: because slow illnesses may violate the exogeneity assumption through anticipation effects, the causal interpretation of this estimate is limited. Nevertheless, the large coefficient (approximately 0.09) and the descriptive finding that coverage becomes more partisan when outlets can anticipate the vacancy are consistent with the theoretical mechanism. These results provide strong support for *H2*, while the evidence speaks against *H3*.

Figure 10 presents the full event-study trajectories for the three main slant specifications across the full observation period. Pre-period coefficients are broadly

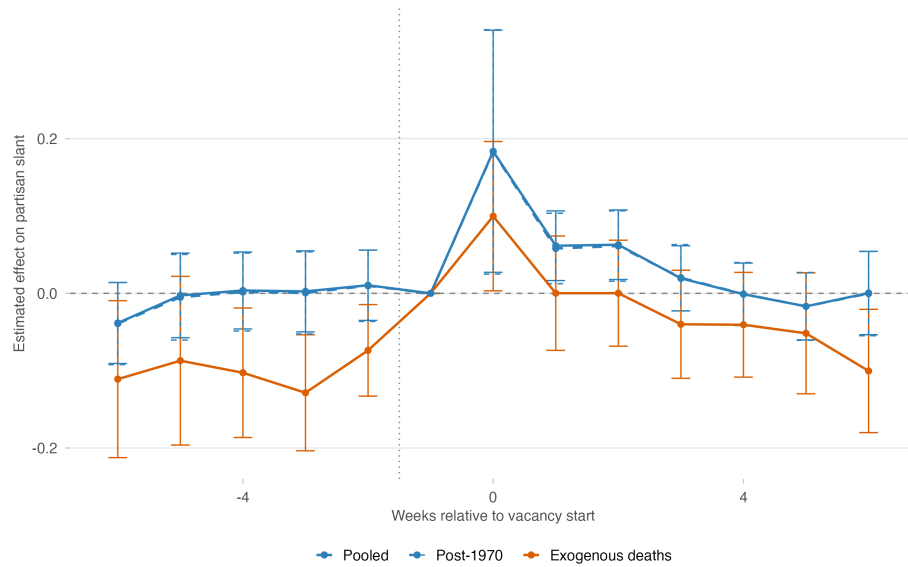


Figure 9: Event study across slant specifications. Pooled, post-1970, and exogenous-deaths estimates overlaid. Error bars show 95% CI clustered by event.

flat, providing visual support for the parallel-trends assumption. The exogenous-deaths specification shows a modest pre-trend in the week immediately preceding the vacancy, which is not unexpected: in cases of sudden fatal illness, media outlets often report on the severity of the condition before the member's death is officially announced. This motivates the donut design reported in the robustness section. Post-period coefficients turn positive at the onset of the campaign window and remain elevated, consistent with a sustained increase in partisan slant during special-election campaigns. The coefficients in the pooled model flatten after a few weeks, which may reflect the diluting influence of non-exogenous vacancies or coincident political events (Formal tests for pre-trends in Appendix B).

4.3 Internal Pluralism

The following results are exploratory given the data limitations documented in the preceding section. Due to limited outlet coverage in the article-level data, pre-trend tests fail across all specifications, and results should be interpreted descriptively rather than causally.

Figure 11 presents event-study estimates for the article-level slant outcome. To examine whether editorial vacancies affect the directional slant and extremeness of individual articles, I construct an article-level stacked panel in which each article appears once per event that covers its publication week. I estimate two-way fixed effects models with *headline slant* and *absolute headline slant* as outcomes. The preferred specification absorbs outlet-by-event fixed effects (outlet:event FE),

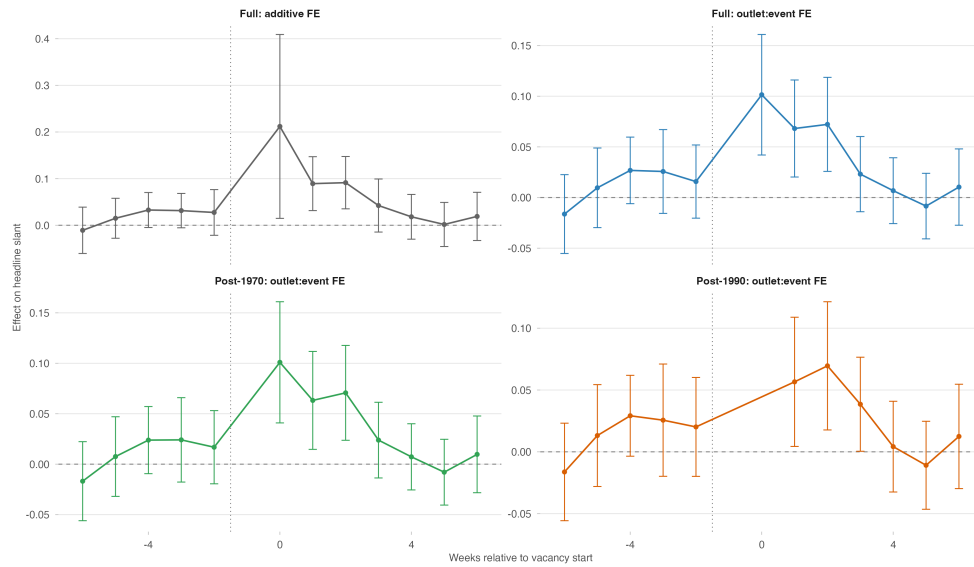


Figure 10: Event study: article-level slant as outcome. Five specifications differing in sample and fixed-effect structure. Blue (Full, outlet $\times$ event FE) is the preferred specification; grey lines show alternatives. Error bars show 95% CI clustered by event.

which eliminates any outlet-specific baseline that may differ across events and identifies the treatment effect purely from within-outlet, within-event variation over time. As a robustness check, I also estimate a model with additive fixed effects for events, outlets, and time periods separately (additive FE), which is less conservative but assumes outlet-specific baselines are constant across events.

The plot reveals two noteworthy patterns. First, the more conservative outlet-event specification shows clearer and stronger effects, providing additional empirical support for *H2*. Second, the post-1990 models suggest diminishing effects over time, a pattern that runs counter to *H3*. Possible explanations include rising journalistic professionalization or intensified competition from emerging media forms, though the difference across periods is not statistically significant and the result should be treated as exploratory given the data limitations noted above (For another operationalization of internal pluralism see Appendix C).

5. Robustness

A key concern in event-study designs is that estimated pre-treatment coefficients may partly reflect anticipation effects rather than a stable baseline. In the context of electoral vacancies caused by illness, journalists and editors may begin adjusting coverage already before the formal vacancy is declared, for instance, as a member's health deteriorates and the prospect of a special election becomes foreseeable. To address this concern, I implement a *donut difference-in-differences* design, which

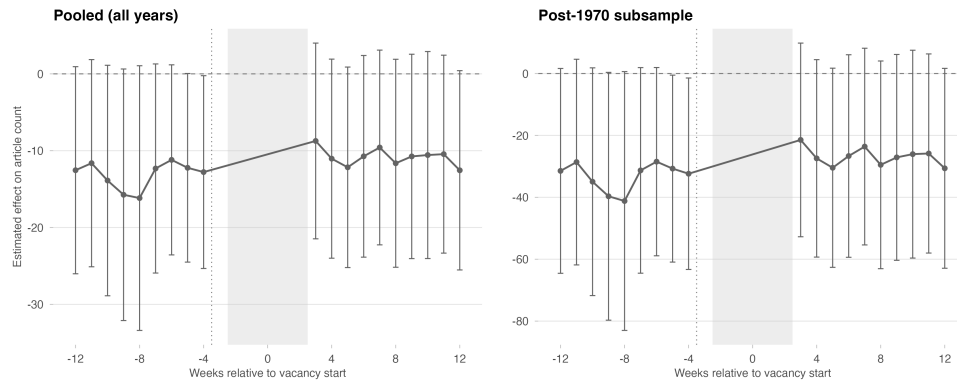


Figure 11: Event study estimates for article volume under the donut design. The grey shaded area marks the excluded period ($t \in \{-2, \dots, 2\}$); the reference period is $t = -3$. Left panel: pooled specification. Right panel: post-1970 subsample.

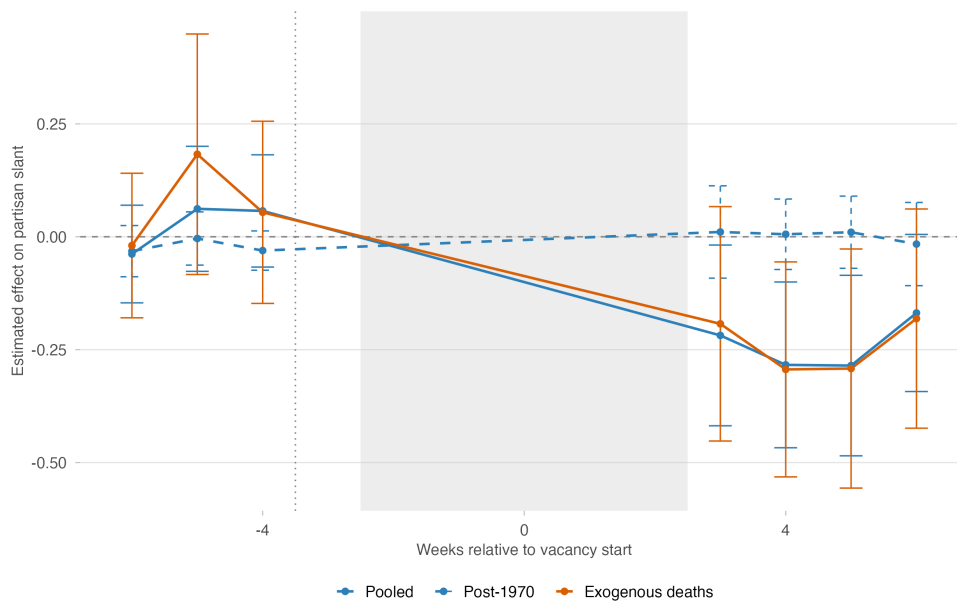


Figure 12: Event study estimates for partisan slant under the donut design, for three specifications. Colors and linetypes distinguish specifications as in the main analysis.

removes the five weeks immediately surrounding the treatment window (event time -2 through $+2$) and shifts the reference period to $t = -3$. This creates a “donut hole” around the transition period, ensuring that identification relies only on variation from weeks further removed from the vacancy start.

Figures 12 and 13 illustrate the event-study coefficients under this specification. The grey shaded band marks the excluded time period. For article volume, the coefficients in the retained pre-period show no systematic pre-trend, and the post-period estimates cannot be distinguished from zero in any specification. For partisan slant, Figure 13 shows broadly flat pre-trends for the pooled and exogenous-deaths specifications. The post-period coefficients shift below zero, indicating a negative effect that is inconsistent in sign with the main results; the post-1970 subsample is the only specification that yields a precisely estimated null effect.

Figure 14 presents the pooled average treatment effects on the treated (ATT) across all specifications and both outcomes under the donut design. For article volume, the point estimates are negative but imprecisely estimated and not statistically significant at conventional levels. For partisan slant, the pooled and exogenous-deaths specifications retain a negative and statistically significant ATT; the effect sizes are even larger than in the main analysis. The post-1970 estimate is close to zero and insignificant. Taken together, the donut results suggest that the slant effects identified in the main analysis are primarily concentrated in the transition period and are not robust across the full treatment window. This could indicate a violation of the assumption of no co-treatment.

6. Discussion

I examined the effect of special election campaigns on the behavior of U.S. print media, constructing an original data set covering all House special elections from 1789 to 2020 and merging it with historical newspaper data. Using stacked difference-in-differences models, I find evidence that special election campaigns increase the partisan slant of headline coverage, though the effect does not appear to be robust to more demanding specifications such as the donut design. Standard theories of political information typically assume that information is generally beneficial for democratic accountability. The results of this study, however, suggest that information in the period leading up to elections may be more partisan than its baseline, with implications for the quality of the information environment available to voters. Whether differential exposure to partisan media environments affects voters’ political attitudes remains an important question for future research.

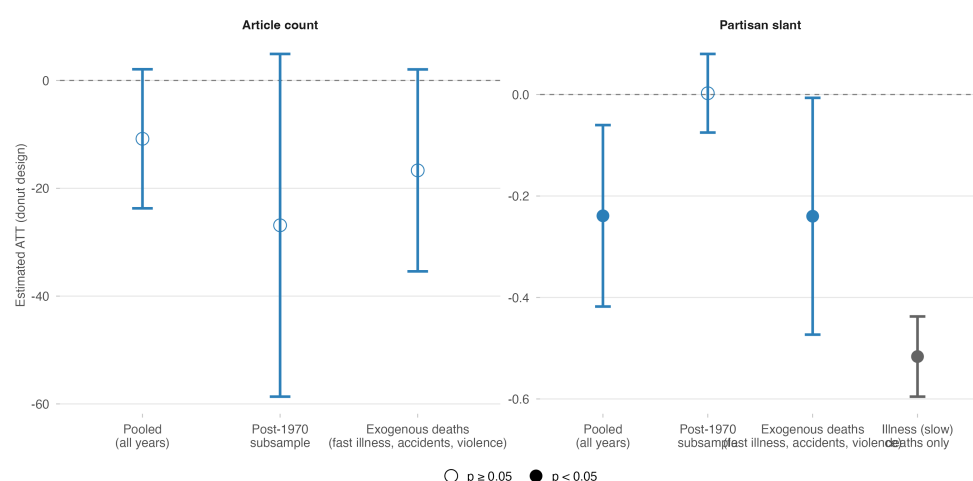


Figure 13: Pooled ATT estimates under the donut design for article volume (left) and partisan slant (right). Filled circles indicate $p < 0.05$; open circles indicate $p \geq 0.05$. Error bars show 95% confidence intervals.

The results should be interpreted with caution for several reasons. First, I use regions to construct treatment and control groups, but even with this strategy, the stable unit treatment value assumption (SUTVA) is likely violated. Historical newspaper outlets often had readerships extending well beyond their immediate region, and major outlets were not consumed exclusively locally. Spillovers of this kind would most likely bias the estimated coefficients toward zero, so the violation is not necessarily fatal to the design, but it limits the precision of the geographic assignment.

A more serious concern relates to the internal validity of the measurement strategy. The language model I use has relatively few parameters and therefore limited ability to understand context, which may introduce systematic classification errors, especially across long historical periods. In addition, I classify only article headlines rather than full article texts. Headlines may be more sensationalist than the underlying content, and this focus may overstate the degree of partisan framing relative to a full-text measure. Additionally, election data about the competitiveness of the special election would increase the internal validity of the concept political intensity.

The most serious concern for the internal validity of my findings is the lack of robustness in the donut-difference-in-difference design. These results indicate, that the result in the main model might be driven by the death of one house-member and not of the election campaign. Future research should focus on identifying the treatment effect of the death and use a donut-design as main specification.

A further limitation concerns the generalizability of the findings. The small

number of newspaper outlets contributes to instability in the estimates. The newspapers in the sample may not be representative of the broader historical media landscape, and a disproportionate share of articles comes from the Chicago Tribune and the New York Times, giving these outlets undue influence over the results.

Moreover the results in this paper are restricted to special elections in the U.S. It remains speculation if the effect would change if for example an incumbent exists or outside of the U.S.

Despite these limitations, the findings are promising for future research. This type of historical media analysis is of substantial substantive relevance, because democratic accountability depends on the quality of information available to voters, and partisan media environments may contribute to both democratic erosion and political polarization.

7. References

- Aalberg, Toril, Jesper Strömbäck, and Claes H. De Vreese. 2012. "The Framing of Politics as Strategy and Game: A Review of Concepts, Operationalizations and Key Findings." *Journalism* 13(2): 162–78. doi:10.1177/1464884911427799.
- Allcott, Hunt, and Matthew Gentzkow. 2017. "Social Media and Fake News in the 2016 Election." *Journal of Economic Perspectives* 31(2): 211–36. doi:10.1257/jep.31.2.211.
- Angrist, Joshua D., and Jörn-Steffen Pischke. 2008. *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton University Press. doi:10.2307/j.ctvc4j72.
- Ashworth, Scott. 2012. "Electoral Accountability: Recent Theoretical and Empirical Work." *Annual Review of Political Science* 15(1): 183–201. doi:10.1146/annurev-polisci-031710-103823.
- Ashworth, Scott, Ethan Bueno De Mesquita, and Amanda Friedenberg. 2017. "Accountability and Information in Elections." *American Economic Journal: Microeconomics* 9(2): 95–138. doi:10.1257/mic.20150349.
- Baumgartner, Frank R., and Bryan D. Jones. 2009. *Agendas and Instability in American Politics*. 2nd ed. Chicago: The University of Chicago Press.
- Bennett, W. Lance. 1990. "Toward a Theory of Press-State Relations in the United States." *Journal of Communication* 40(2): 103–27. doi:10.1111/j.1460-2466.1990.tb02265.x.
- Bennett, W. Lance, Regina G. Lawrence, and Steven Livingston. 2008. *When the press fails: political power and the news media from Iraq to Katrina*. Chicago, Ill.: University of Chicago Press.
- Blumler, Jay G., and Michael Gurevitch. 1995. *The Crisis of Public Communication*. London; New York: Routledge.
- Bode, Leticia, David S. Lassen, Young Mie Kim, Dhavan V. Shah, Erika Franklin Fowler, Travis Ridout, and Michael Franz. 2016. "Coherent Campaigns? Campaign Broadcast and Social Messaging." *Online Information Review* 40(5): 580–94. doi:10.1108/OIR-11-2015-0348.
- Campbell, Angus, Philip E. Converse, Miller E. Warren, and Donald E. Stokes. 1960. *The American Voter*. University of Chicago Press.
- Cappella, Joseph N., and Kathleen Hall Jamieson. 1997. *Spiral of Cynicism: The Press and the Public Good*. New York: Oxford University Press.
- Chadwick, Andrew. 2017. *The Hybrid Media System: Politics and Power*. Second Edition. New York, NY: Oxford University Press.
- Cook, Timothy E. 2005. *Governing with the News: The News Media as a Political Institution*. 2nd ed. Chicago: University of Chicago Press.

- Cottle, Simon, ed. 2000. *Ethnic Minorities and the Media: Changing Cultural Boundaries*. Buckingham; Philadelphia: Open University Press.
- Dimitrova, Daniela V, and Jesper Strömbäck. 2012. "Election News in Sweden and the United States: A Comparative Study of Sources and Media Frames." *Journalism* 13(5): 604–19. doi:10.1177/1464884911431546.
- Downs, Anthony. 1957. "An Economic Theory of Democracy." *Harper and Row* 28.
- Druckman, James N., Erik Peterson, and Rune Slothuus. 2013. "How Elite Partisan Polarization Affects Public Opinion Formation." *American Political Science Review* 107(1): 57–79. doi:10.1017/S0003055412000500.
- Fiorina, Morris P. 1978. "Economic Retrospective Voting in American National Elections: A Micro-Analysis." *American Journal of Political Science* 22(2): 426. doi:10.2307/2110623.
- Fowler, Erika Franklin. 2022. "Political Advertising in the United States; Second Edition."
- Gans, Herbert J. 2004. *Deciding what's news: a study of CBS evening news, NBC nightly news, Newsweek, and Time*. 25th anniversary edition. Evanston, Ill: Northwestern University Press.
- Gelman, Andrew, and Gary King. 1993. "Why Are American Presidential Election Campaign Polls So Variable When Votes Are So Predictable?"
- Gentzkow, Matthew, and Jesse M Shapiro. 2010. "What Drives Media Slant? Evidence From U.S. Daily Newspapers." *Econometrica* 78(1): 35–71. doi:10.3982/ECTA7195.
- Gentzkow, Matthew, and Jesse M. Shapiro. 2006. "Media Bias and Reputation." *Journal of Political Economy* 114(2): 280–316. doi:10.1086/499414.
- Gerber, Alan S., Dean Karlan, and Daniel Bergan. 2009. "Does the Media Matter? A Field Experiment Measuring the Effect of Newspapers on Voting Behavior and Political Opinions." *American Economic Journal: Applied Economics* 1(2): 35–52. <http://www.jstor.org/stable/25760159> (March 18, 2026).
- Hallin, Daniel C., and Paolo Mancini. 2004. *Comparing media systems: three models of media and politics*. Cambridge: Cambridge University Press.
- Hamilton, James T. 2004. *All the News That's Fit to Sell: How the Market Transforms Information into News*. Princeton University Press.
- Highton, Benjamin, and Cindy D. Kam. 2011. "The Long-Term Dynamics of Partisanship and Issue Orientations." *The Journal of Politics* 73(1): 202–15. doi:10.1017/S0022381610000964.
- Hirano, Shigeo, and James M. Snyder, Jr. 2009. "Using Multimember District Elections to Estimate the Sources of the Incumbency Advantage." *American*

- Journal of Political Science* 53(2): 292–306. doi:10.1111/j.1540-5907.2009.00371.x.
- Huddy, Leonie, Lilliana Mason, and Lene Aarøe. 2015. “Expressive Partisanship: Campaign Involvement, Political Emotion, and Partisan Identity.” *American Political Science Review* 109(1): 1–17. doi:10.1017/S0003055414000604.
- Iyengar, Shanto, and Kyu S Hahn. 2009. “Red Media, Blue Media: Evidence of Ideological Selectivity in Media Use.” *Journal of Communication* 59(1): 19–39. doi:10.1111/j.1460-2466.2008.01402.x.
- Iyengar, Shanto, and Donald R. Kinder. 2010. *News That Matters: Television and American Opinion*. Updated ed. Chicago: University of Chicago Press.
- Iyengar, Shanto, and Adam F. Simon. 2000. “New Perspectives and Evidence on Political Communication and Campaign Effects.” *Annual Review of Psychology* 51(1): 149–69. doi:10.1146/annurev.psych.51.1.149.
- Kaplan, Richard L. 2003. “Politics and the American Press: The Rise of Objectivity, 1865–1920.” *Canadian Journal of Communication* 28(2). doi:10.22230/cjc.2003v28n2a1370.
- Layman, Geoffrey C., Thomas M. Carsey, and Juliana Menasce Horowitz. 2006. “PARTY POLARIZATION IN AMERICAN POLITICS: Characteristics, Causes, and Consequences.” *Annual Review of Political Science* 9(1): 83–110. doi:10.1146/annurev.polisci.9.1.83.
- Lazarsfeld, Paul F., Bernard Berelson, and Hazel Gaudet. 2021. *The People’s Choice: How the Voter Makes up His Mind in a Presidential Campaign*. Legacy edition. New York: Columbia University Press.
- Lelkes, Yphtach, Gaurav Sood, and Shanto Iyengar. 2017. “The Hostile Audience: The Effect of Access to Broadband Internet on Partisan Affect.” *American Journal of Political Science* 61(1): 5–20. doi:10.1111/ajps.12237.
- “List of Members of the United States Congress Who Died in Office (1900–1949).” 2026. *Wikipedia*. [https://en.wikipedia.org/w/index.php?title=List_of_members_of_the_United_States_Congress_who_died_in_office_\(1900%E2%80%931949\)&oldid=1344160707](https://en.wikipedia.org/w/index.php?title=List_of_members_of_the_United_States_Congress_who_died_in_office_(1900%E2%80%931949)&oldid=1344160707) (March 19, 2026).
- “List of Special Elections to the United States House of Representatives.” 2026. *Wikipedia*. https://en.wikipedia.org/w/index.php?title=List_of_special_elections_to_the_United_States_House_of_Representatives&oldid=1343384589 (March 19, 2026).
- Lodge, Milton, and Charles S. Taber. 2013. *The rationalizing voter*. Cambridge: Cambridge University Press.
- Mason, Lilliana. 2016. “A Cross-Cutting Calm: How Social Sorting Drives Affective Polarization.” *Public Opinion Quarterly* 80(S1): 351–77. doi:10.1093/poq/nfw001.
- Mason, Lilliana. 2018. *Uncivil agreement: how politics became our identity*. Chicago London: The University of Chicago Press.

- McCombs, Maxwell E., and Donald L Shaw. 1972. "THE AGENDA-SETTING FUNCTION OF MASS MEDIA."
- McCombs, Maxwell E., and Sebastián Valenzuela. 2021. *Setting the agenda: the news media and public opinion*. Third edition. Cambridge, UK Medford, MA: Polity.
- Messing, Solomon, and Sean J. Westwood. 2014. "Selective Exposure in the Age of Social Media: Endorsements Trump Partisan Source Affiliation When Selecting News Online." *Communication Research* 41(8): 1042–63. doi:10.1177/0093650212466406.
- Patterson, Thomas E. 2011. *Out of Order: An Incisive and Boldly Original Critique of the News Media's Domination of Ameri*. Westminster: Knopf Doubleday Publishing Group.
- Petrova, Maria. 2011. "Newspapers and Parties: How Advertising Revenues Created an Independent Press." *American Political Science Review* 105(4): 790–808. doi:10.1017/S0003055411000360.
- Prior, Markus. 2007. *Post-broadcast democracy: how media choice increases inequality in political involvement and polarizes elections*. Cambridge: Cambridge University Press. doi:10.1017/CBO9781139878425.
- Roth, Jonathan, Pedro H.C. Sant'Anna, Alyssa Bilinski, and John Poe. 2023. "What's Trending in Difference-in-Differences? A Synthesis of the Recent Econometrics Literature." *Journal of Econometrics* 235(2): 2218–44. doi:10.1016/j.jeconom.2023
- Schudson, Michael. 2001. "The Objectivity Norm in American Journalism*." *Journalism* 2(2): 149–70. doi:10.1177/146488490100200201.
- Shoemaker, Pamela J., and Timothy Vos. 2009. *Gatekeeping Theory*. 0th ed. Routledge. doi:10.4324/9780203931653.
- Snyder, James M., and David Strömberg. 2010. "Press Coverage and Political Accountability." *Journal of Political Economy* 118(2): 355–408. doi:10.1086/652903.
- Strömberg, David. 2004. "Mass Media Competition, Political Competition, and Public Policy." *The Review of Economic Studies* 71(1): 265–84. doi:10.1111/0034-6527.00284.
- Strömberg, David. 2015. "Media and Politics." *Annual Review of Economics* 7(1): 173–205. doi:10.1146/annurev-economics-080213-041101.
- Stroud, Natalie Jomini. 2011. *Niche news: the politics of news choice*. Oxford: Oxford University Press. doi:10.1093/acprof:oso/9780199755509.001.0001.
- Taber, Charles S., and Milton Lodge. 2006. "Motivated Skepticism in the Evaluation of Political Beliefs." *American Journal of Political Science* 50(3): 755–69. doi:10.1111/j.1540-5907.2006.00214.x.
- Tuchman, Gaye. 1972. "Objectivity as Strategic Ritual: An Examination of

- Newsman's Notions of Objectivity." *American Journal of Sociology* 77(4): 660–79. doi:10.1086/225193.
- Vreese, Claes H. 2005. "News Framing: Theory and Typology." *Information Design Journal* 13(1): 51–62. doi:10.1075/idjdd.13.1.06vre.
- Zaller, John. 1999. *A Theory of Media Politics: How the Interests of Politicians, Journalists, and Citizens Shape the News*. University of Chicago Press.
- Zaller, John R. 1992. *The Nature and Origins of Mass Opinion*. 1st ed. Cambridge University Press. doi:10.1017/CBO9780511818691.

8. Notes

The replication materials for the full paper can be found here: <https://github.com/NiklasHaehn/historical-pluralism>

9. Appendix

9.1 A. Measurement Specifications

9.1.1 TDM Studio Search

In TDM Studio, I restricted the articles to English-language news articles in which the full content mentions “congress, senate, election, campaign, representative, senator”. Specifically, I am using the following prompt: *LA(english) AND FULL-TEXT(congress OR senate OR election OR campaign OR representative OR senator)*

9.1.2 LLM Prompt for slant

I used the following prompt:

” You are a political text annotation assistant.

Task: Classify the partisan framing of this U.S. newspaper headline.

Return ONE of: - D = headline favors Democrats / the liberal side, or criticizes Republicans / the conservative side - R = headline favors Republicans / the conservative side, or criticizes Democrats / the liberal side - N = neutral or balanced — no clear partisan slant - NA = not a U.S. political headline, or insufficient context to determine partisan framing

Definitions: - D: The wording explicitly portrays Democrats, liberals, or the left more favorably, or portrays Republicans, conservatives, or the right more negatively (e.g., loaded praise for Democrats, blame language toward Republicans). - R: The wording explicitly portrays Republicans, conservatives, or the right more favorably, or portrays Democrats, liberals, or the left more negatively. - N: Neutral, factual, or symmetric reporting. No evaluative slant toward either side. Includes straightforward reporting of political events, votes, or statements without loaded language. - NA: The headline is not about U.S. politics, involves no partisan actors, or is too vague or incomplete to classify.

Rules: 1. Use only the headline text. Do not use outside knowledge. 2. Evaluate explicit wording only — do not infer unstated meanings or assume bias from context. 3. Coverage of only one party is NOT automatically partisan. Return D or R only if the wording itself is evaluative or loaded. If coverage is neutral and factual, return N even if only one party is mentioned. 4. If uncertain between D and R, return N. 5. Return only one value: D, R, N, or NA.

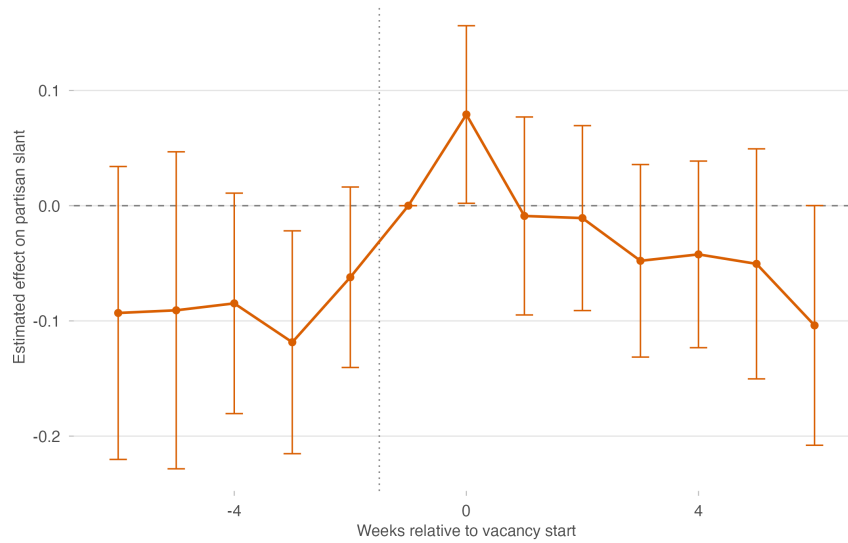


Figure 14: Event study for partisan slant: illness (fast) deaths only. Error bars show 95% CI clustered by event.

Headline: {headline} ”

9.2 B. Robustness: Illness (fast) Subsample

Figure B1 replicates the main slant event study restricting the sample to special elections triggered by fast-onset illnesses only. This is the most conservative exogeneity restriction: these deaths were, by definition, not anticipated by the media environment. The pre-period is flat and the post-period pattern mirrors the main result, providing strong support for the causal interpretation.

9.3 B. HonestDiD Sensitivity Analysis

Figure B2 presents the Roth et al. (2023) sensitivity analysis for each of the four main DiD specifications: pooled, post-1970, exogenous deaths, and illness (slow). The horizontal axis shows $\bar{M}$, the maximum pre-trend violation as a multiple of the largest observed pre-period coefficient. The shaded bands display the robust confidence set under each violation level. A specification is considered robust if the confidence set excludes zero for small values of $\bar{M}$.

9.4 C. Internal Pluralism: SD of Headline Slant

Figure C3 shows the event-study estimates for internal pluralism, operationalized as the standard deviation of article-level headline slant within an outlet-week. A higher within-outlet standard deviation indicates more heterogeneous coverage. The figure is presented for two outlets with sufficient article density across the observation period (NYT, Chicago Tribune). Pre-trend tests fail, so these results

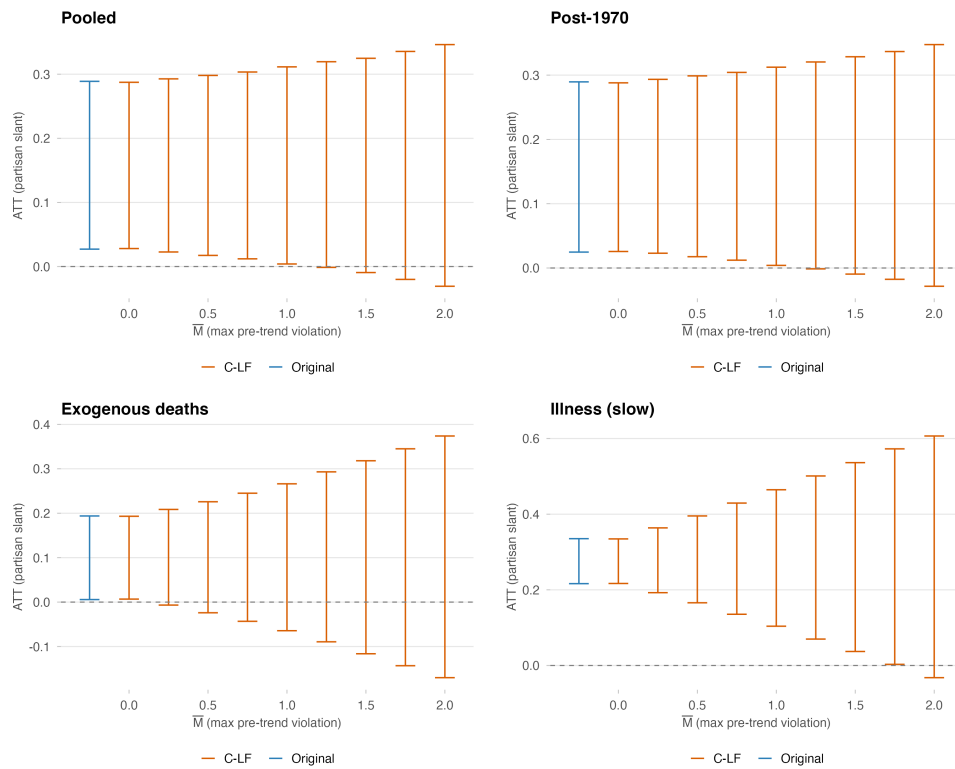


Figure 15: HonestDiD sensitivity analysis. Each panel corresponds to one DiD specification. Shaded areas show robust confidence sets as $\bar{M}$ increases from 0 (standard CI) to 2.

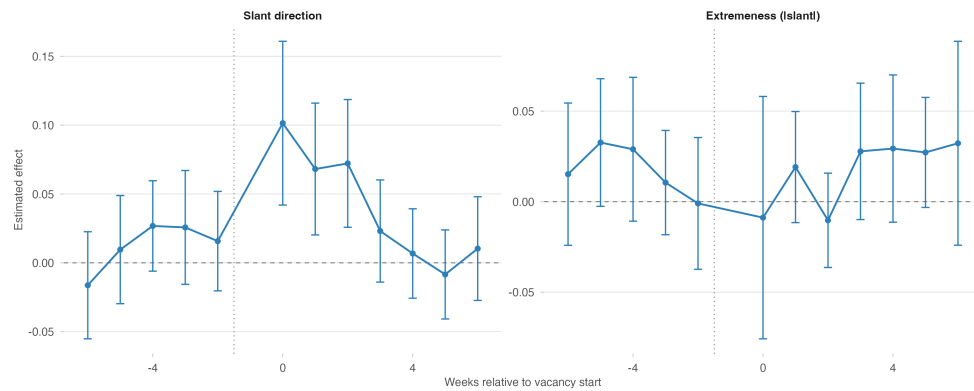


Figure 16: Event study: internal media pluralism (SD of headline slant). Pre-trends fail; results are descriptive. NYT and Chicago Tribune only.

should be interpreted as descriptive evidence only.